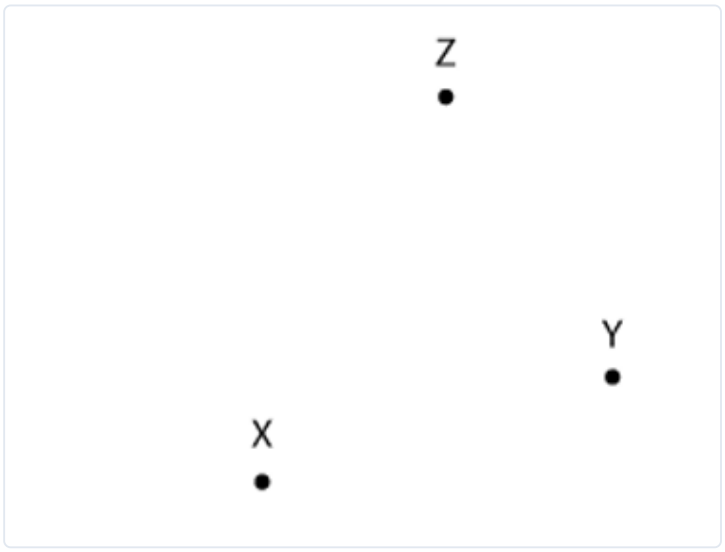
CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

2.1 Equations of Motion

Syllabus Section: 2. Kinematics **Paper Type:** Multiple Choice (Paper 1) **Total:** 32 Questions (32 Marks)

Question 1 Easy [1 mark]

An object moves directly from X to Z



In a shorter time, a second object moves from X to Y to Z.

Which statement about the two objects is correct for the journey from X to Z?

A they have the same average speed

B they have the same average velocity

C they have the same displacement

D they travel the same distance

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Displacement describes an objects change in position, measuring from its starting position to the final position
- The objects both finish at the same position so despite taking different routes, they have the same displacement

A is incorrect because the distance travelled by the second object is greater than the first one but it reaches Z in a shorter time, since average speed = total distance / total time its average speed should be greater

B is incorrect because the time taken for the second object to reach Z is less than the first and they both have the same displacement, since average velocity = displacement / time the second object has a greater velocity

D is incorrect because the first object travels directly from X to Z which is a shorter distance than X to Y to Z

Question 2 Easy [1 mark]

An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

A the height of fall and the time of fall

B the height of fall and the weight of the body

C the mass of the body and the height of fall

D the mass of the body and the time of fall

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- To determine the measurements needed to determine the acceleration of free fall, we will need to consider the equation
  - $s = u \quad t \quad + \frac{1}{2} a \quad t^2$
- We need to outline the quantities that we know
  - $s = h$ (height)
  - $a = g$ (acceleration of free fall)
  - t (time)
- Since the falling body will initially be at rest, i.e. u = 0 we can simplify this to
  - $h = \frac{1}{2} \quad g \quad t^2$
- Rearranging the equation above we can see
  - $g = \frac{2h}{t^2}$
- So, the only measurements required are the height and time of the fall

Question 3 Easy [1 mark]

Which feature of a graph allows acceleration to be determined?

A the area under a displacement-time graph

B the area under a velocity-time graph

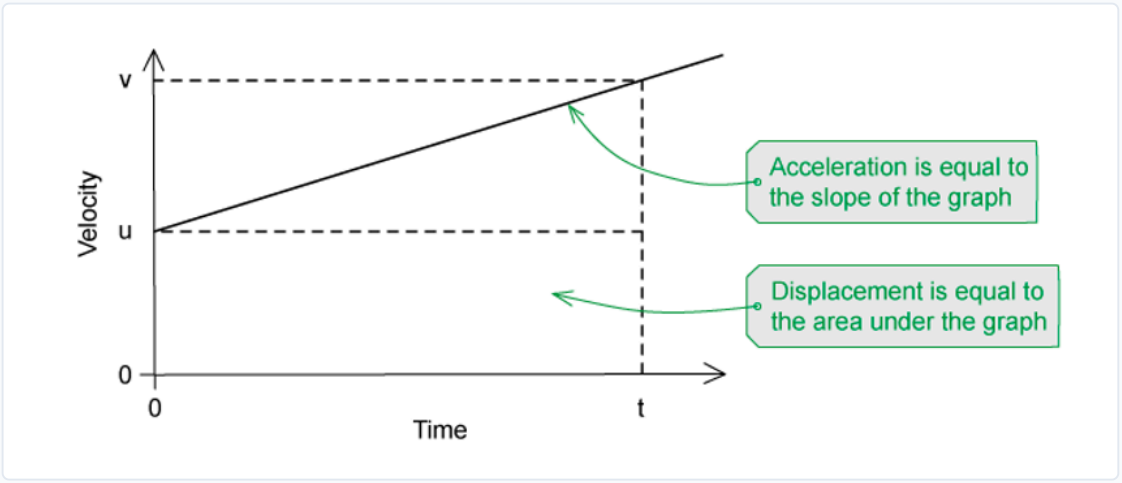
C the slope of a displacement-time graph

D the slope of a velocity-time graph

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Acceleration is equal to change in velocity / time
- Looking at a velocity-time graph we can see $\frac{\Delta y}{\Delta x}$ is equal to $\frac{v-u}{t}$ which is equivalent to $\frac{change\ \ in\ \ velocity}{time}$
- Therefore, acceleration is equal to the slope of the graph



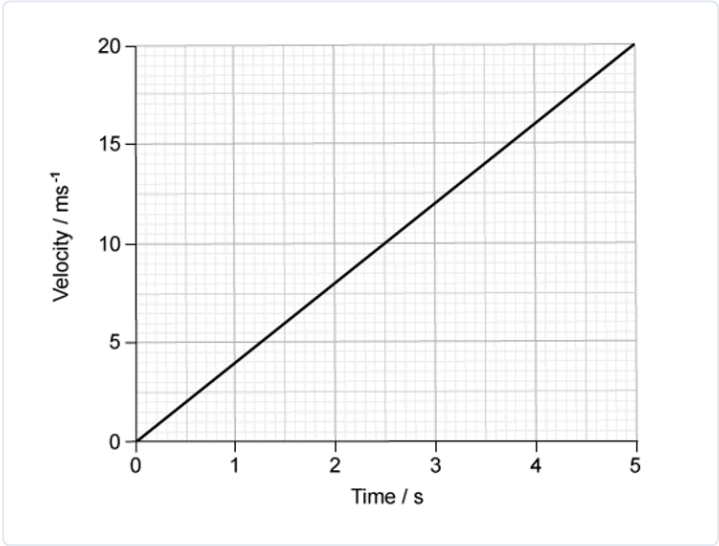
A is incorrect because the area under a displacement-time graph is displacement × time which is not a physical quantity that has any use in physics

B is incorrect because the area under a velocity-time graph is equal to displacement since displacement = velocity × time

C is incorrect because the slope of a displacement-time graph is equal to velocity since velocity = displacement / time

Question 4 Easy [1 mark]

The velocity of an object during the first five seconds of its motion is shown on the graph



What is the displacement of the object in this time?

A 4 m

B 20 m

C 50 m

D 100 m

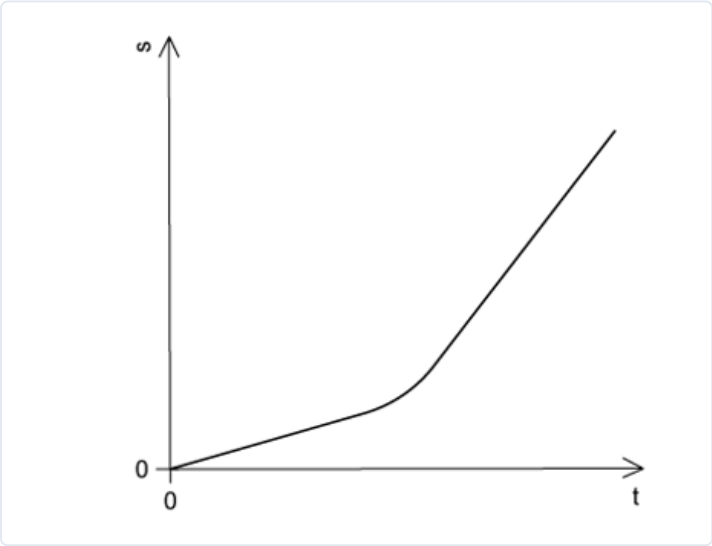
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Displacement is equal to velocity × time
- This can be found from the area under the velocity - time graph
- Area under the graph = ½ × base × height = ½ × Δ v × Δ t
- Displacement = ½ × 20 × 5 = 50 m

Question 5 Easy [1 mark]

The variation with time *t* of the displacement *s* moved by a body is shown below.



What can be deduced from the graph about the motion of the body?

A it accelerates continuously

B it starts from rest

C the displacement is proportional to time

D the speed changes

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- There are two clear sections on this displacement-time graph
- At first the object moves with a constant velocity and at a later time it accelerates briefly until reaching a higher velocity, which it continues to travel at for the remainder of the distance

A is incorrect because if it was accelerating continuously, the line would curve upwards. Instead, it travels constantly at two different velocities

B is incorrect because it starts from a velocity which is equal to the gradient of the line, if it started from rest, the line would be horizontal and curve upwards to show an acceleration

C is incorrect because the displacement is proportional to time for the sections of the graph which are straight but it is not proportional the entire time

Question 6 Easy [1 mark]

An object is thrown with velocity 7.1 m s^{-1} vertically upwards on the Moon. The acceleration due to gravity on the Moon is 1.62 m s^{-2}

What is the time taken for the object to return to its starting point?

A 3.5 s

B 4.4 s

C 6.5 s

D 8.8 s

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- First it is a good idea to write down all the quantities that we know
  - Initial velocity, u = 7.1 m s^{-1}
  - a = 1.62 m s^{-2}
  - the final velocity, v = 0 m s^{-1} (at the maximum height)
  - t = *?*
- The equation that relates *u, v, a* and *t* is
  - $v = u + a\ t$
  - $v - u = a\ t$
  - $t = \frac{v-u}{a} = \frac{7.1}{1.62} = 4.4$ s
- It takes 4.4 s for the object to reach its maximum height, and it will take the same time to reach the ground since the acceleration is the same, and the final velocity will be the same, so:
  - Total time = 4.4 × 2 = 8.8 s

Question 7 Easy [1 mark]

A ball is thrown upwards with an initial vertical velocity of 2.2 m s^{-1}.

What is the maximum height reached by the ball?

A 10 cm

B 25 cm

C 50 cm

D 75 cm

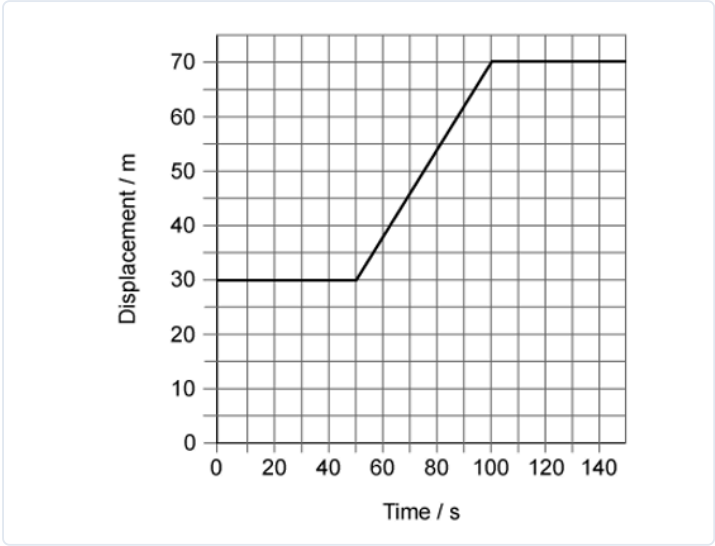
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Write down all the known information:
  - Initial velocity, u = 2.2 m s^{-1}
  - At the maximum height, the final velocity is v = 0 m s^{-1}
  - As the ball moves upwards, $a = -g$ = -9.81 m s^{-2}
  - Maximum height, s = ?
- The equation which relates *u, v, s* and *a* is
  - $v^2 = u^2 + 2as$
- Rearranging for *s* gives:
  - $2as = v^2 - u^2$
  - $s = \frac{v^2 - u^2}{2a}$
  - $s = \frac{2.2^2 - 0}{2 \times 9.81} = 0.25 \text{ m}$
- This is equivalent to 25 cm

Question 8 Easy [1 mark]

A car at rest in a traffic queue moves forward in a straight line and then comes to rest again. The graph shows how the car's displacement varies with time



What is its speed while it is moving?

A 0.70 m s^{-1}

B 0.80 m s^{-1}

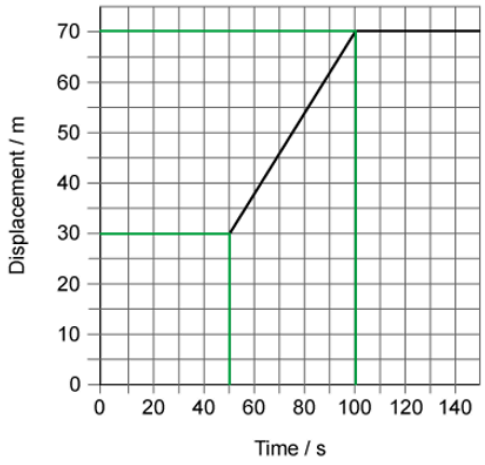
C 1.25 m s^{-1}

D 1.40 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Speed is equal to $\frac{displacement}{time}$ which is given by the gradient of the graph $\frac{\Delta y}{\Delta x}$
- Between 0 – 50 s and 100 – 140 s, when the lines are horizontal, the car is stationary
- Between 50 – 100 s the car is moving with constant speed and is displaced by 70 – 30 = 40 m
- The speed when the car is moving is equal to $\frac{40\ m}{50\ s}$ = 0.8 m s^{-1}


Displacement / m
70
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
Time / s

Question 9 Easy [1 mark]

On a particular railway, a train driver applies the brake of the train at a yellow signal, a distance of 1.0 km from a red signal, where the train stops.

The maximum deceleration of the train is 1.1 m s^{-2}

Assuming uniform deceleration, what is the maximum safe speed of the train at the yellow signal?

A 2.2 m s^{-1}

B 20 m s^{-1}

C 47 m s^{-1}

D 220 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- First, we must write down the variables that we know:
  - s = 1000 m
  - a = –1.1 m s^{-2}
  - v = 0
  - u = ?
- The equation that links *s, a, v* and *u* is:
  - $v^2 = u^2 + 2as$

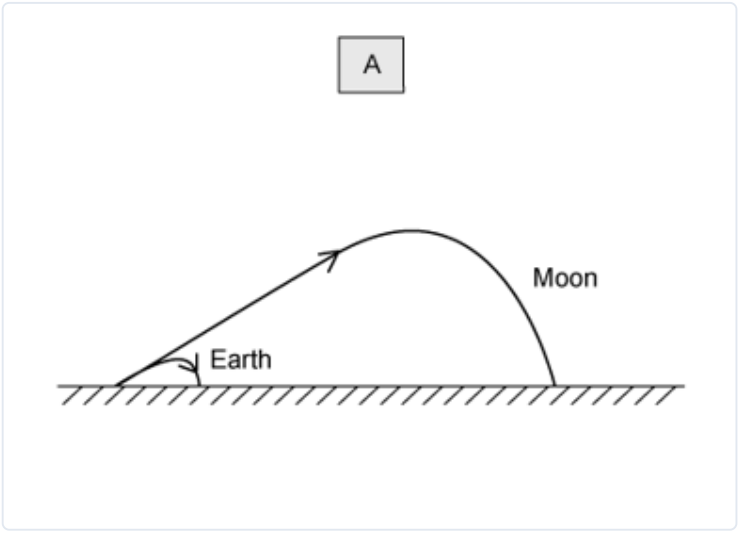
  Rearrange for *u* and calculate:
  - $u^2 = v^2 - 2as$
  - $u^2 = 0 - 2 \times -1.1 \times 1000 = 2200$
  - $u = \sqrt{2200} = 46.9 = 47$ m s^{-1} (2 s.f.)

Question 10 Easy [1 mark]

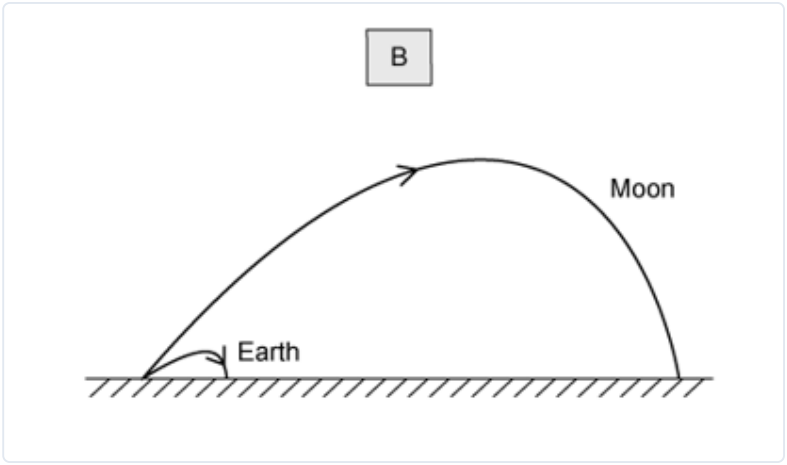
A golf ball is hit with the same force and direction on the Earth and on the Moon.

Which diagram best represents the shapes of the paths taken by the golf ball?

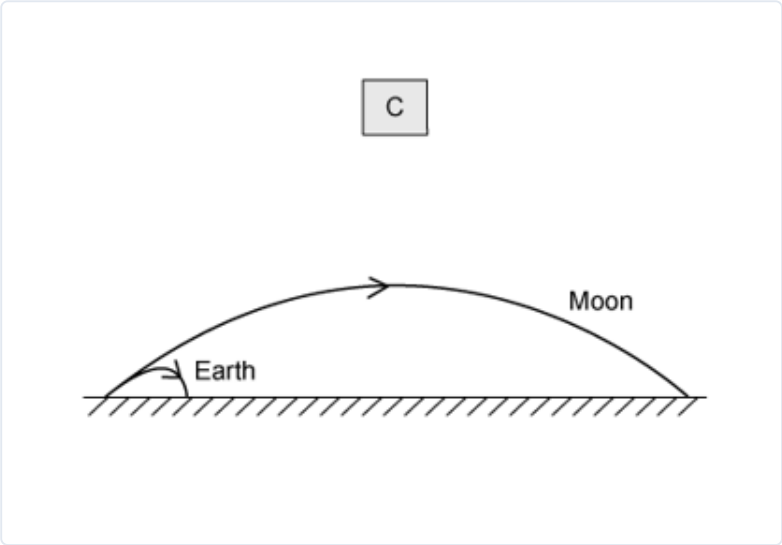
A



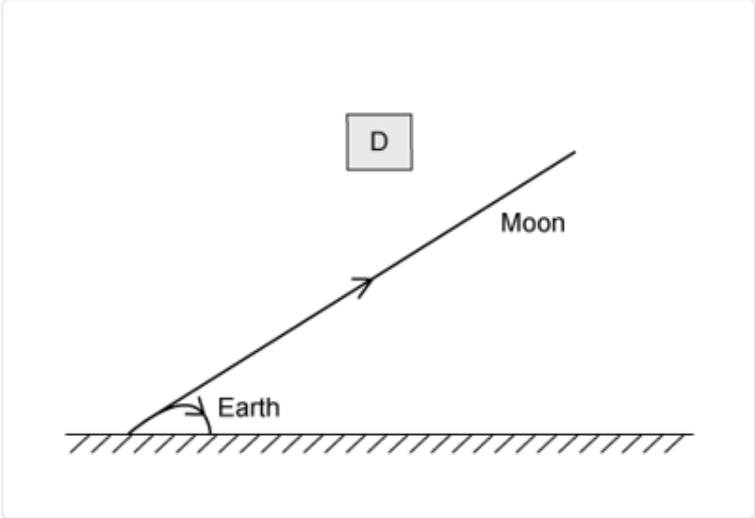
B



C



D



Mark Scheme & Explanation:

*The correct answer is **C** because:*

- When the golf ball is struck upwards, the force of gravity will always bring it back to the surface
- The force of gravity is much higher on the Earth so the golf ball is pulled towards the surface faster which is why it has a much shorter range than the ball on the Moon
- The shape of the golf ball's path depends on air resistance – on Earth, gravity pulls the ball down while air resistance opposes its motion leading to a path which is not symmetrical
- On the Moon, however, there is no atmosphere, so there is no air resistance acting on the ball. This means the path of the golf ball will be symmetrical, as shown in C

A & B are incorrect because these paths are asymmetrical which implies there is an opposing force acting on the ball, such as air resistance, but since there is no atmosphere on the Moon, this is not possible

D is incorrect because the gravity on the Moon is small but it is not zero, so the ball would return to the surface

Question 11 Easy [1 mark]

Which feature of a graph allows velocity to be determined?

A the area under a displacement-time graph

B the area under a velocity-time graph

C the slope of a displacement-time graph

D the slope of a velocity-time graph

Mark Scheme & Explanation:

The correct answer is C

- Velocity is equal to change in displacement / time
- Looking at a displacement-time graph we can see

$$\frac{\Delta y}{\Delta x} = \frac{\Delta s}{t} = \frac{\text{change in displacement}}{\text{time}}$$

- Therefore, velocity is equal to the slope of the graph

A is incorrect because the area under a displacement-time graph is displacement × time which is not a physical quantity that has any use in physics

B is incorrect because the area under a velocity-time graph is equal to displacement since displacement = velocity × time

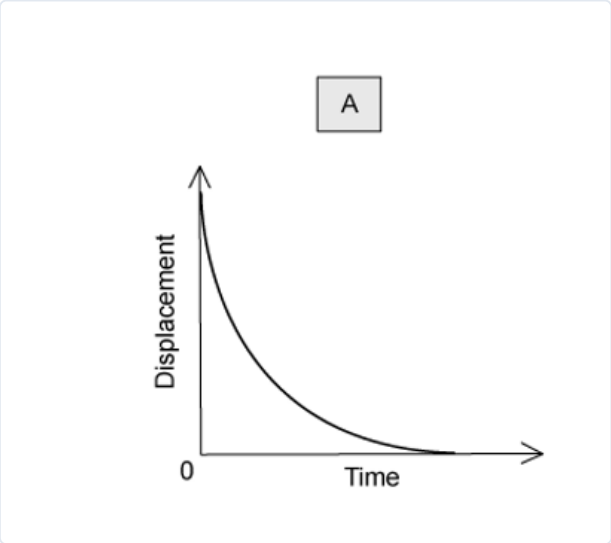
D is incorrect because the slope of a velocity-time graph is equal to acceleration since acceleration = change in velocity / time

Question 12 Medium [1 mark]

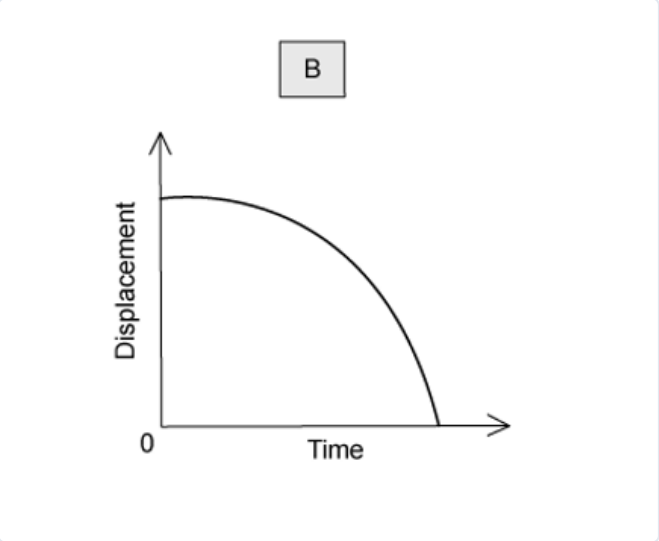
A sphere is released and falls. Its initial acceleration reduces until it eventually begins to travel at constant terminal velocity.

Which displacement-time graph best represents the motion of the sphere?

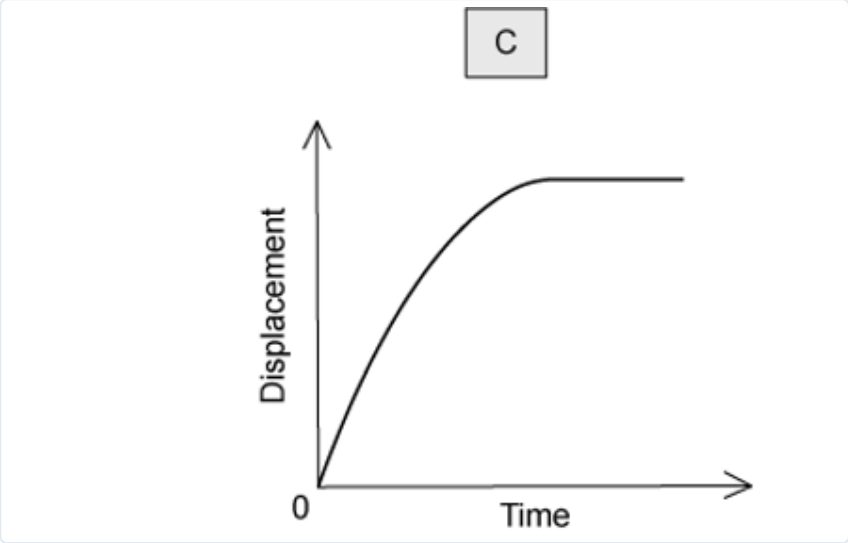
A



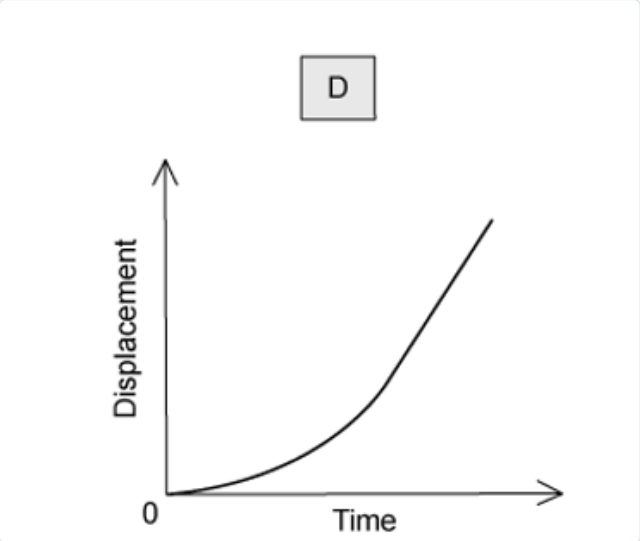
B



C



D



Mark Scheme & Explanation:

*The correct answer is **D** because:*

- As the ball falls further from its starting position, the displacement increases meaning the gradient increases
- Terminal velocity is the maximum velocity reached by the falling sphere, meaning the first section of the graph should curve upwards to show the initial acceleration and the second section should be a straight line to show the constant velocity

A & B are incorrect because the gradients of these graphs are negative

C is incorrect because the horizontal straight line means the velocity is zero and the ball has stopped moving which is not the case

Question 13 Medium [1 mark]

A bullet is fired horizontally with speed *v* from a rifle. For a short time *t* after leaving the rifle, the only force affecting its motion is gravity. The acceleration of free fall is *g*.

Which expression gives the value of $\frac{\text{horizontal distance travelled in time } t}{\text{vertical distance travelled in time } t}$?

A $\frac{vt}{g}$

B $\frac{v}{gt}$

C $\frac{2vt}{g}$

D $\frac{2v}{gt}$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The acceleration of freefall is downwards and has no effect on horizontal motion
- We can find the horizontal distance travelled using
  - distance = speed × time
  - $s_H = vt$ **(1)**
- Since the bullet is fired in a horizontal direction, the vertical component of the velocity is initially zero
- We can find the vertical distance using
  - $s = ut + \frac{1}{2}at^2$
- $u = 0$ and $a = g$, so
  - $s_V = \frac{1}{2}gt^2$ **(2)**
- Combining equations **(1)** and **(2)** leads to
  - $\frac{\text{horizontal distance travelled in time } t}{\text{vertical distance travelled in time } t} = \frac{s_H}{s_V}$
  - $\frac{s_H}{s_V} = \frac{vt}{\frac{1}{2}gt^2} = \frac{2v}{gt}$

Question 14 Medium [1 mark]

The acceleration of free fall on the Moon is one-sixth of that on Earth. On Earth it takes time *t* for a stone to fall from rest a distance of 2 m.

What is the time taken for a stone to fall from rest a distance of 2 m on the Moon?

A $6t$

B $\frac{t}{6}$

C $t\sqrt{6}$

D $\frac{t}{\sqrt{6}}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

Acceleration of free fall on Earth = g

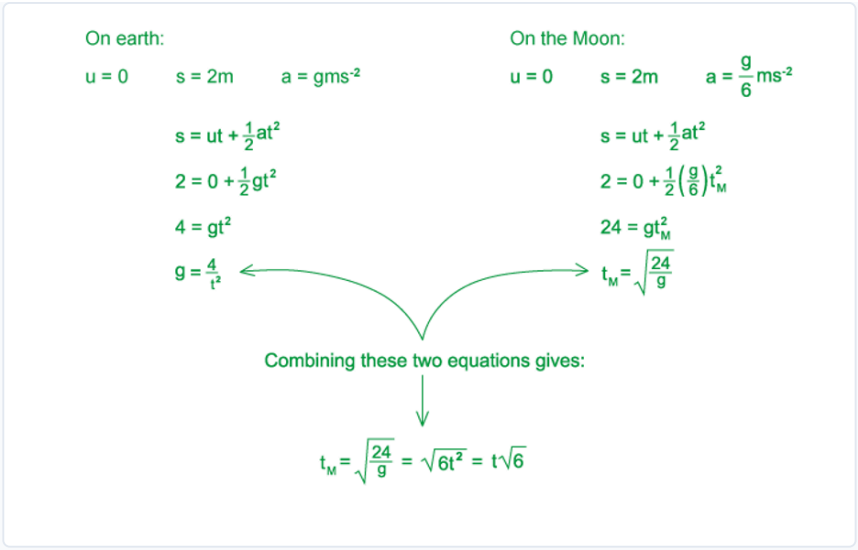


Acceleration of free fall on the Moon = $\frac{g}{6}$



The equation that links s, u, t and a is:

$s = ut + \frac{1}{2}at^2$


On earth:
u = 0 s = 2m a = gms-2
s = ut + ½at²
2 = 0 + ½gt²
4 = gt²
g = 4/t²
On the Moon:
u = 0 s = 2m a = g/6 ms-2
s = ut + ½at²
2 = 0 + ½(g/6)t²M
24 = gt²M
tM = √(24/g)
Combining these two equations gives:
tM = √(24/g) = √6t² = t√6

Question 15 Medium [1 mark]

A stone is thrown upwards and follows a curved path.



Air resistance is negligible.

Why does the path have this shape?

A the stone has a constant horizontal velocity and constant vertical acceleration

B the stone has a constant horizontal acceleration and constant vertical velocity

C the stone has a constant upward acceleration followed by a constant downward acceleration

D the stone has a constant upward velocity followed by a constant downward velocity

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The horizontal and vertical components of velocity and acceleration can be treated independently
- In the absence of air resistance, the horizontal component of velocity is constant while the vertical component of velocity downwards increases at a constant acceleration of 9.81 m s^{-2} (due to the force of gravity)

B & D are incorrect because once the stone has been thrown, it is no longer being acted on by the force that propelled it into the air so there is no horizontal acceleration. The force of gravity causes the stone to decelerate upwards and accelerate downwards at a constant rate meaning the vertical velocity is not constant

C is incorrect because the stone has a constant upward **deceleration** followed by a constant downward acceleration

Question 16 Medium [1 mark]

A train passes through a station at a constant speed of 20 m s^{-1}

A second train is at rest at the station. It begins to leave the station with a uniform acceleration of 1.0 m s^{-2} just as the first train goes past.

How much time passes before the second train overtakes the first train?

A 5 s

B 10 s

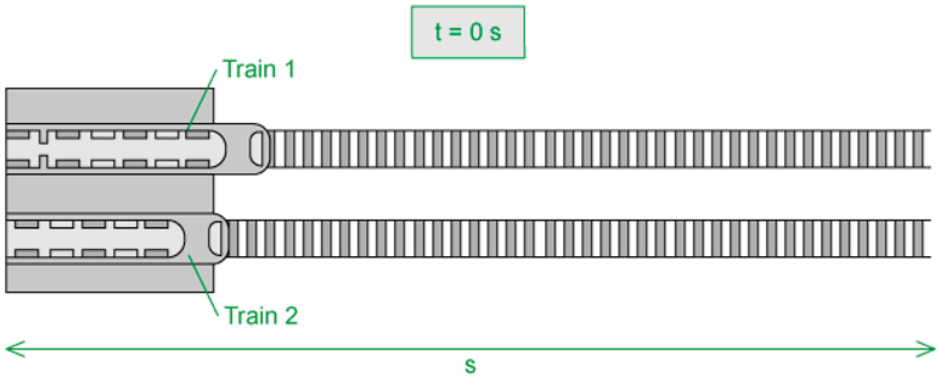
C 20 s

D 40 s

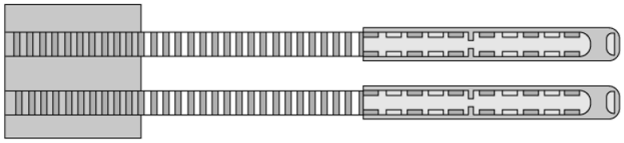
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Train 2 leaves the station as train 1 passes, so the station can be taken as the origin for the time and distance
- For train 2 to overtake train 1, the distance, *s*, travelled by both trains must be equal and the time, *t*, is how long it takes for this to happen



t = ? s



Train 1

Train 1 travels at a constant speed of 20 ms^{-1}

distance = speed × time

s = 20t

Train 2

a = 1.0 ms^{-2} u = 0 ms^{-1} s = 20 t t = ?

The equation that links these quantities is:

$s = ut + \frac{1}{2}at^2$

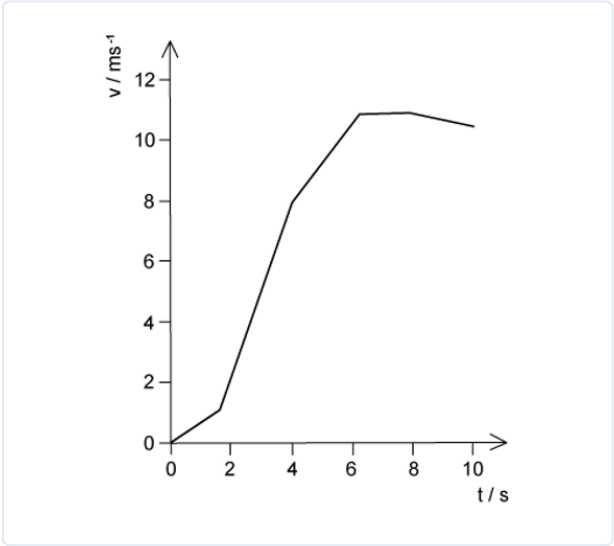
$20t = 0 + \frac{1}{2}(1)t^2$

$40t = t^2$

t = 40 s

Question 17 Medium [1 mark]

The graph shows how the speed *v* of a sprinter changes with time *t* during a 100 m race.



What is the best estimate of the maximum acceleration of the sprinter?

A 3 m s^{-2}

B 6 m s^{-2}

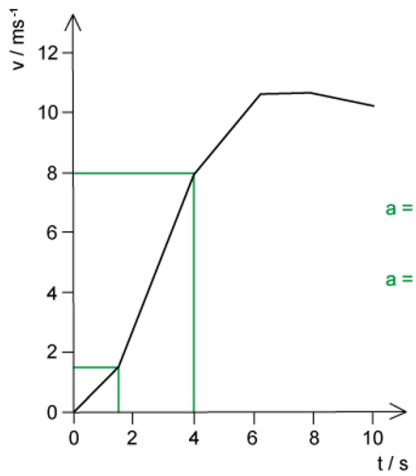
C 9 m s^{-2}

D 12 m s^{-2}

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Acceleration is represented by the gradient of the velocity-time graph
- The steeper the gradient the higher the acceleration, so to find the maximum acceleration we need to estimate the gradient of the steepest line

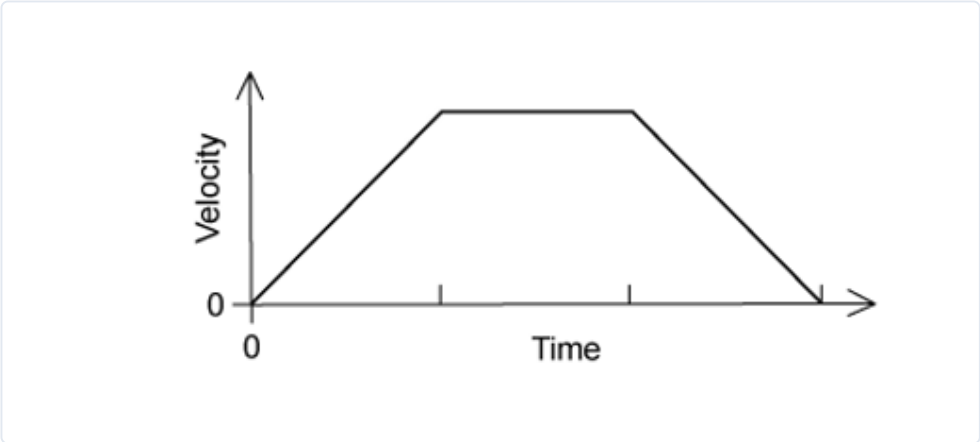


$$a = \frac{\Delta v}{\Delta t}$$

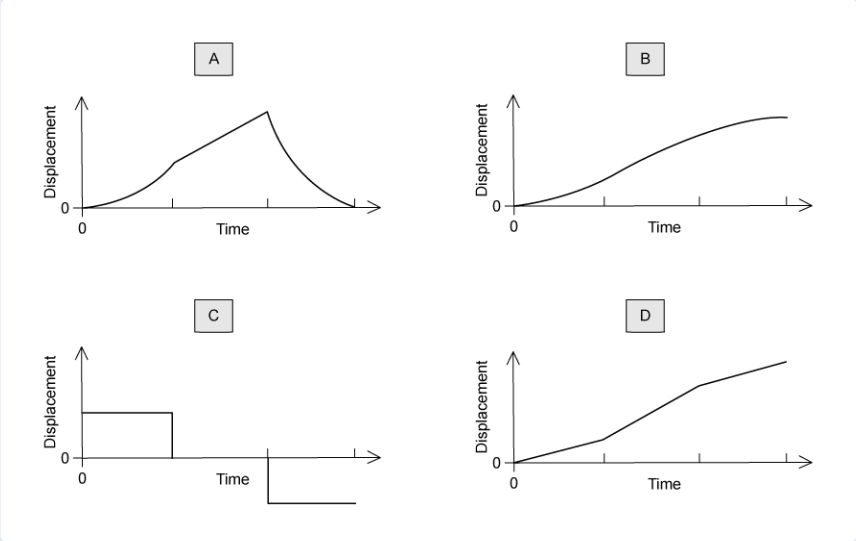
$$a = \frac{8 - 1.2}{4.1 - 1.8} = \frac{6.8}{2.3} \approx 3\ ms^{-2}$$

Question 18 Medium [1 mark]

The graph of velocity against time for an object moving in a straight line is shown



What is the corresponding graph of displacement against time?



A

B

C

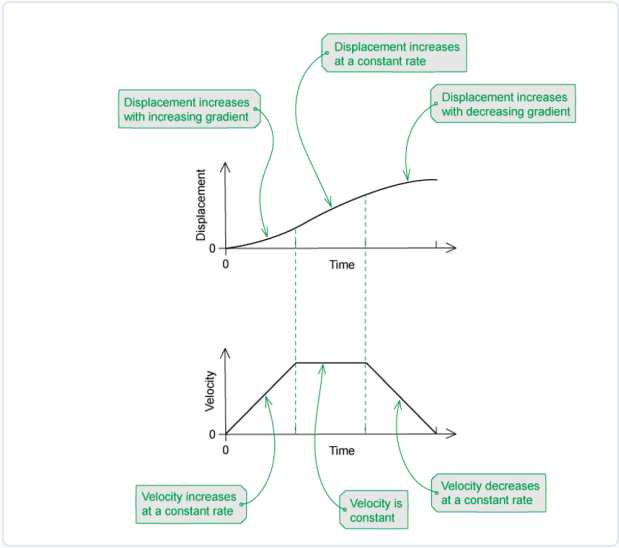
D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The gradient of the velocity-time graph gives the acceleration and the area under the graph gives the displacement
- The velocity is always positive, so the displacement is always positive

- The gradient of the displacement-time graph gives the velocity



Question 19 Medium [1 mark]

In a cathode-ray tube, an electron is accelerated uniformly in a straight line from a speed of 4×10^5 m s^{-1} to one tenth of the speed of light over a distance of 10 mm

What is the acceleration of the electron?

A 4.5×10^3 m s^{-2}

B 4.5×10^6 m s^{-2}

C 4.5×10^{13} m s^{-2}

D 4.5×10^{16} m s^{-2}

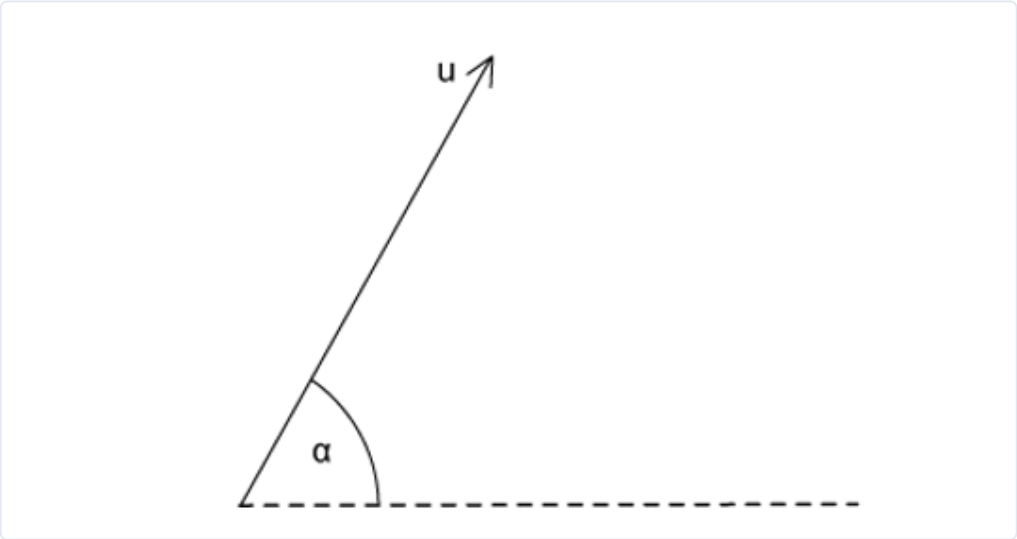
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- First let's write out the quantities we have been given
  - $u = 4 \times 10^5$ m s^{-1}
  - $s = 10$ mm $= 10 \times 10^{-3}$ m
  - $v = 3 \times 10^8 \times \frac{1}{10} = 3 \times 10^7$ m s^{-1}
  - a = ?
- The equation that links *u, v, s* and *a* is
  - $v^2 = u^2 + 2as$
- We need to rearrange this to find *a*
  - $2as = v^2 - u^2$
  - $a = \frac{v^2 - u^2}{2s}$
  - $a = \frac{3 \times 10^{7\,2} - 4 \times 10^{5\,2}}{2 \times 10 \times 10^{-3}} = 4.5 \times 10^{16}$ m s^{-2}

Question 20 Medium [1 mark]

A projectile is fired at an angle α to the horizontal at a speed u, as shown.



What are the vertical and horizontal components of its velocity after a time t?

Assume that air resistance is negligible. The acceleration of free fall is g.

	Vertical component	Horizontal component
A	$u \sin \alpha$	$u \cos \alpha$
B	$u \sin \alpha - gt$	$u \cos \alpha - gt$
C	$u \sin \alpha - gt$	$u \cos \alpha$
D	$u \cos \alpha$	$u \sin \alpha - gt$

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

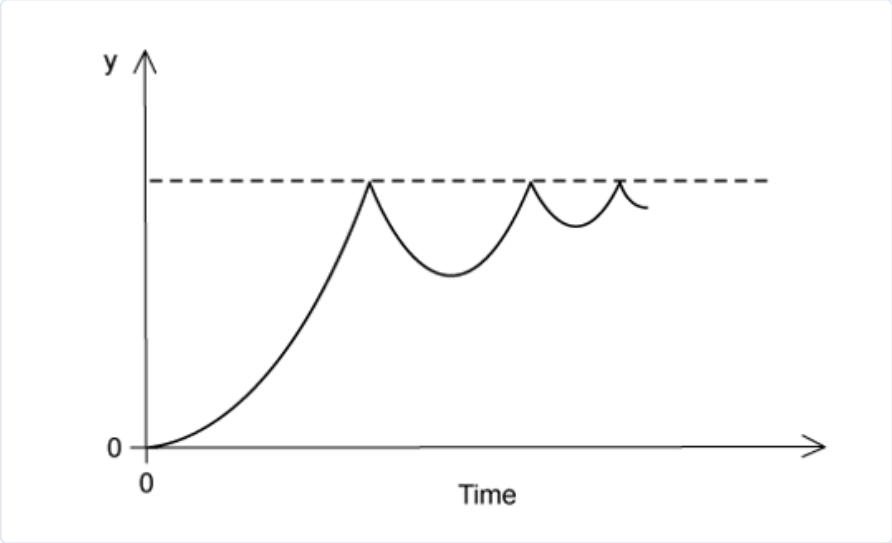
- We can use trigonometry to deduce
  - Horizontal (→) component = $u \cos \alpha$
  - Vertical (↑) component = $u \sin \alpha$

- Because of gravity, we have to consider the free fall of acceleration, which is equal to g
- This only affects the vertical component of velocity, and acts in the opposite direction to the motion
- So, at a time, t, the vertical component of velocity = $u \sin \alpha - gt$
- The horizontal component is not influenced by gravity so is equal to $u \cos \alpha$

Question 21 Medium [1 mark]

A ball is released from rest above a horizontal surface and bounces several times.

The graph shows how, for this ball, a quantity *y* varies with time.



What is the quantity *y*?

A acceleration

B displacement

C kinetic energy

D velocity

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- If we take the point when the ball is at rest as the origin and the horizontal surface as its maximum displacement
- The graph shows an increasing and decreasing gradient as the ball accelerates and decelerates with each bounce, which is indicative of a displacement-time graph
- Each time the ball strikes the surface, it loses some energy through sound and heat, which is why its displacement decreases each time but does not come to zero as it does not reach its original height

A is incorrect because the acceleration is constant, so the lines would be straight, not curved

C & D are incorrect because when the ball hits the surface, its velocity at that moment is zero. So, its kinetic energy at that moment would also be zero. The graph does not reduce to zero, so y cannot be kinetic energy or velocity

Question 22 Medium [1 mark]

A spacecraft is launched from Earth's surface and accelerates uniformly in a straight line from an initial speed of 2×10^3 m s^{-1} to a final speed of 6×10^3 m s^{-1} over a distance of 400 km.

What is the acceleration of the spacecraft?

A 4×10^{-3} m s^{-2}

B 4×10^{1} m s^{-2}

C 4×10^{3} m s^{-2}

D 4×10^{6} m s^{-2}

Mark Scheme & Explanation:

The correct answer is B

- List the known quantities:
  - $u = 2 \times 10^3$ m s^{-1}
  - $s = 400$ km $= 400 \times 10^3$ m
  - $v = 6 \times 10^3$ m s^{-1}
  - a = ?
- The equation that links *u*, *v*, *s*, and *a* is

$$v^2 = u^2 + 2as$$

- Rearrange to find *a*

$$2as = v^2 - u^2$$

$$a = \frac{v^2 - u^2}{2s}$$

- Substitute in known values to calculate *a*

$$a = \frac{6 \times 10^{3\,2} - 2 \times 10^{3\,2}}{2 \times 400 \times 10^3}$$

$$a = 40 \text{ m s}^{-2}$$

$$a = 4 \times 10^1 \text{ m s}^{-2} \text{ 1 s.f.}$$

Question 23 Hard [1 mark]

A body having uniform acceleration *a* increases its velocity from *u* to *v* in time *t*.

Which expression would **not** give a correct value for the body's displacement during time *t* ?

A $ut + \frac{1}{2}at^2$

B $vt - \frac{1}{2}at^2$

C $\frac{v+uv-u}{2a}$

D $\frac{v-ut}{2}$

Mark Scheme & Explanation:

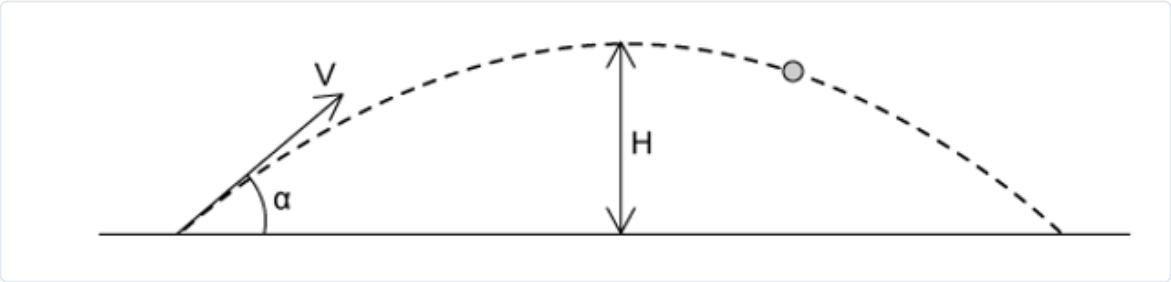
*The correct answer is **D** because:*

- $Average\ \ speed\ \ =\ \ \frac{v+u}{2}$
- $Distance\ \ =\ \ speed\ \times\ time\ \ =\ \ \frac{v+ut}{2}$
- D would be correct if it had a plus sign

A is incorrect as	this equation is equal to displacement $s = ut + \frac{1}{2}at^2$
B is incorrect as	if we combine the equations $s = ut + \frac{1}{2}at^2 \ \ and \ \ v = u + at$ $u = v - at$ $s = v - att + \frac{1}{2}at^2$ $s = vt - at^2 + \frac{1}{2}at^2$ $s = vt - \frac{1}{2}at^2$
C is incorrect as	if we take the equation $v^2 = u^2 + 2as$ $2as = v^2 - u^2$ $s = \frac{v^2-u^2}{2a}$ we can factorise $v^2 - u^2 = v + uv - u$ so, $\frac{v+uv-u}{2a}$

Question 24 Hard [1 mark]

A cannon fires a cannonball with an initial speed *v* at an angle α to the horizontal.



Which equation is correct for the maximum height H reached?

A $H = \frac{v \sin \alpha}{2g}$

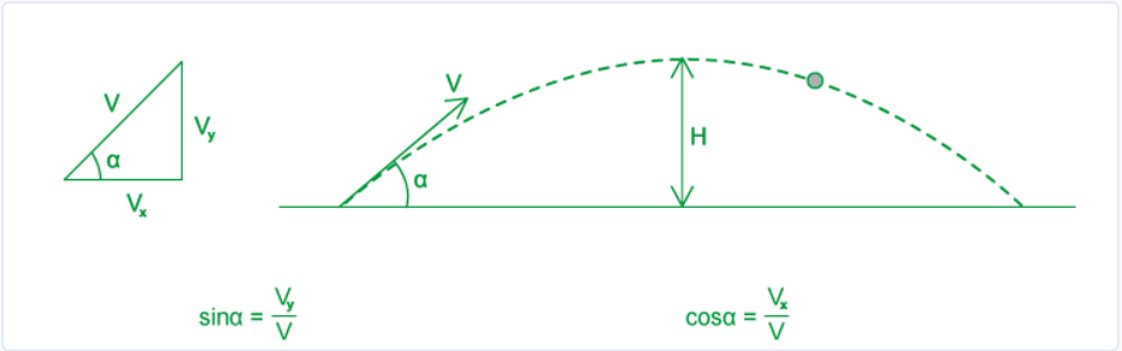
B $H = \frac{g \sin \alpha}{2v}$

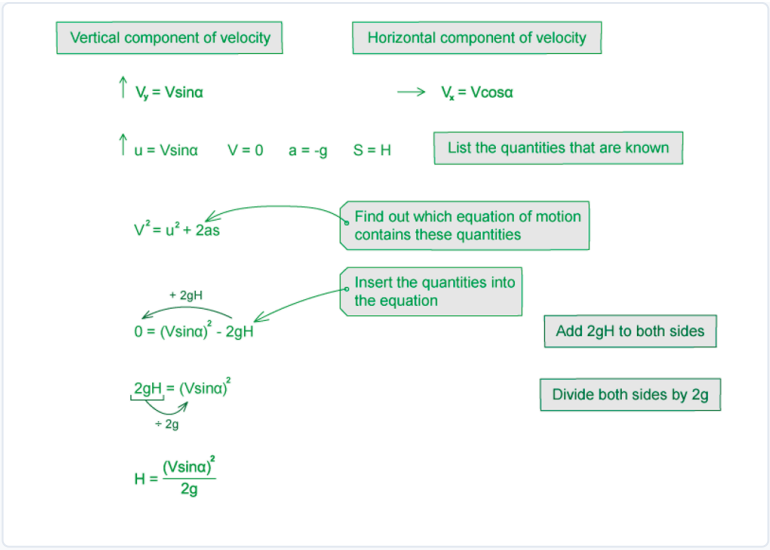
C $H = \frac{v \sin \alpha^2}{2g}$

D $H = \frac{g^2 \sin \alpha}{2v}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*



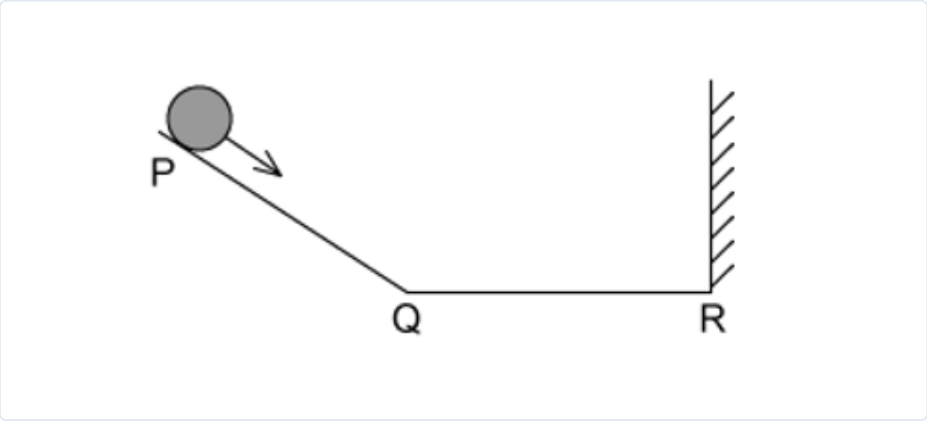

Vertical component of velocity
Horizontal component of velocity
↑ $V_y = V\sin\alpha$
→ $V_x = V\cos\alpha$
↑ u = Vsinα V = 0 a = -g S = H
List the quantities that are known
$V^2 = u^2 + 2as$
Find out which equation of motion contains these quantities
Insert the quantities into the equation
+ 2gH
$0 = (V\sin\alpha)^2 - 2gH$
Add 2gH to both sides
$2gH = (V\sin\alpha)^2$
÷ 2g
Divide both sides by 2g
$H = \frac{(V\sin\alpha)^2}{2g}$

Question 25 Hard [1 mark]

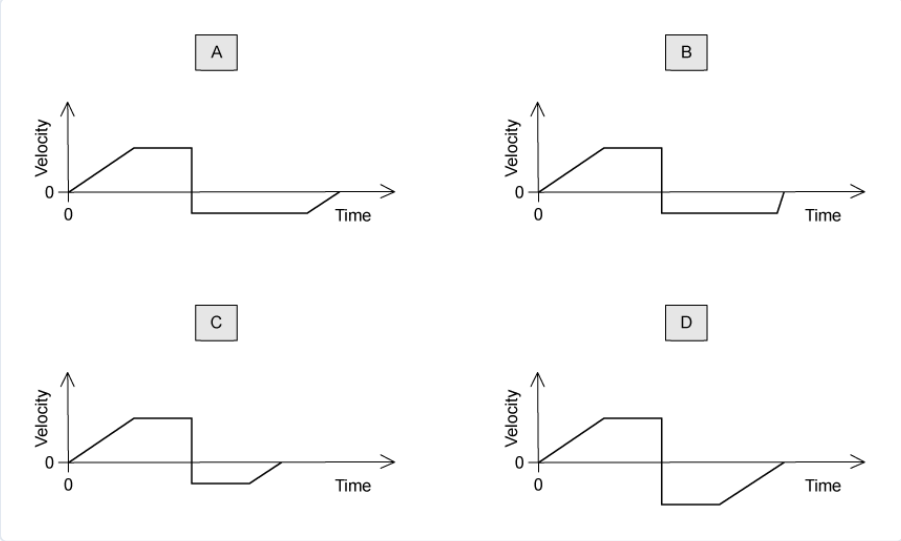
A ball is released from rest on a smooth slope PQR.

It moves down the slope, along a smooth horizontal surface QR and rebounds inelastically at R.

Then it moves back to Q and comes to rest momentarily somewhere on PQ.



Which velocity-time graph represents the motion of the ball?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- From P to Q, the weight of the ball causes it to accelerate uniformly. Therefore, the increase in speed is uniform so the gradient is constant

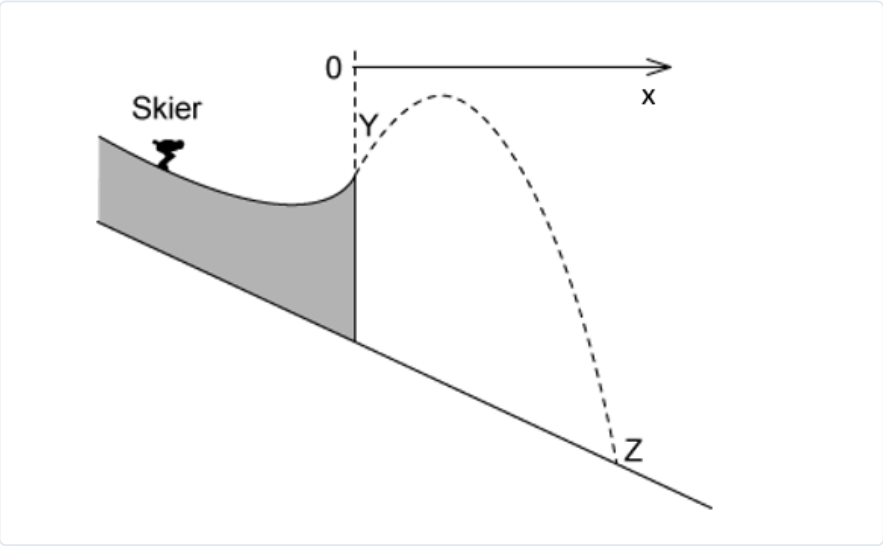
- From Q to R, the motion of the ball is horizontal, so the acceleration of free fall does not affect its motion which is shown on the graph as a horizontal line
- After the collision with the wall, the kinetic energy decreases since the collision is inelastic, this results in a decrease in speed (KE = ½ mv^2)
- Therefore, the ball will take longer to return to the slope, meaning the negative horizontal velocity should be longer

B is incorrect because the gradient of the graph in the final section is too steep. Since the gradient represents the acceleration due to gravity, it should be the same as the gradient in the first section.

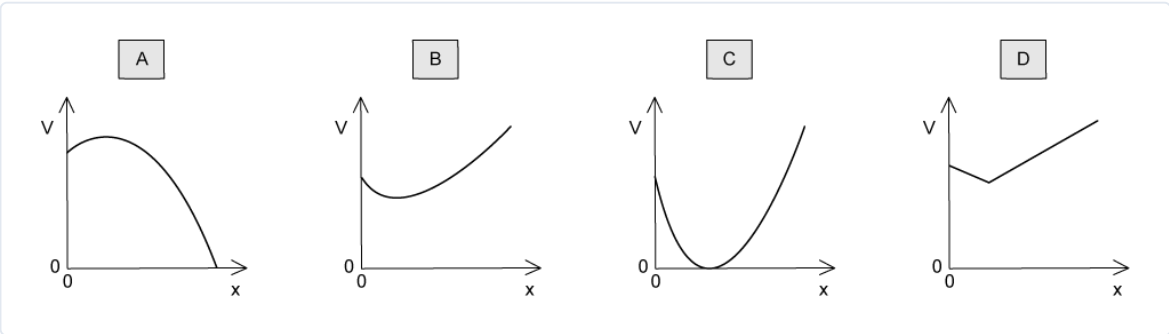
C & D are incorrect because the negative horizontal velocities on these graphs are too short, and the value of the negative velocity in graph D has not decreased.

Question 26 Hard [1 mark]

The dotted line shows the path of a competitor in a ski-jumping competition.



Ignoring air resistance, which graph best represents the variation of his speed *v* with the horizontal distance *x* covered from the start of his jump at Y before landing at Z?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- At the skier's maximum height, the speed is at a minimum since gravitational potential is at a maximum and kinetic energy is at a minimum
- The speed does not reach zero since it still has horizontal velocity
- Just before landing, the speed is not zero. Since Y is higher than Z, the person has both potential energy and kinetic energy at Y

- Because of the conservation of energy, kinetic energy is higher at Z so the speed is higher here

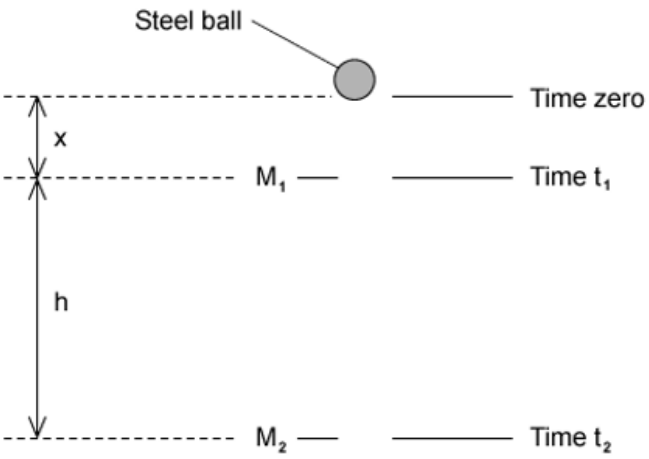
A is incorrect because this graph shows the skier has maximum kinetic energy (highest *v*) and minimum gravitational energy at the top of the path, which is the opposite to what really happens

C is incorrect because the speed doesn't reach zero as the skier still has horizontal velocity at the top of the path

D is incorrect because the change in *v* and *x* is not uniform due to the acceleration due to gravity

Question 27 Hard [1 mark]

Two markers M_1 and M_2 are set up a vertical distance h apart.



A steel ball is released at time zero from a point a distance x above M_1. The ball reaches M_1 at time t_1 and reaches M_2 at time t_2. The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

A $\frac{2h}{t_2^2}$

B $\frac{2h}{t_1 + t_2}$

C $\frac{2h}{t_2^2 - t_1^2}$

D $\frac{2h}{t_2 - t_1^2}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- From the origin to marker M_1:
- Initially the ball is at rest, so:
  - $u = 0\ m\ s^{-1}$
  - $t = t_1$
  - $s = x$

- The equation that links *s, u, t* and *a* is
  - $s = ut + \frac{1}{2}at^2$
  - $x = 0 + \frac{1}{2}at_1{}^2$
  - $x = \frac{1}{2}at_1{}^2$ **(1)**
- From the origin to marker M_2:
  - u = 0 m s^{-1}
  - t = t_2
  - s = x + h
- Using the same equation as before:
  - $s = \quad ut + \frac{1}{2}at^2$
  - $x + h = 0 + \frac{1}{2}at_2{}^2$
  - $x + h = \frac{1}{2}at_2{}^2$ **(2)**
- If we take equation **(1)** from **(2)**:
  - $x + h - x = \frac{1}{2}at_2{}^2 - \frac{1}{2}at_1{}^2$
  - $h = \frac{1}{2}at_2{}^2 - t_1{}^2$
  - $a = \frac{2h}{t_2{}^2 - t_1{}^2}$

Question 28 Hard [1 mark]

A sprinter runs a 200 m race in a straight line. He starts from rest and accelerates at 2.5 m s^{-2} until reaching a maximum speed of 10 m s^{-1}. He maintains this speed until he crosses the finish line.

How long does it take the sprinter to run the race?

A 8 s

B 20 s

C 22 s

D 40 s

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- There are two parts to this motion
  - Motion with constant acceleration of 2.5 m s^{-2}
  - Motion with constant speed of 10 m s^{-1}
- If the duration of the acceleration is t_1 then we want to use an equation to find this, where: $t = t_1$, a = 2.5 m s^{-2}, u = 0 m s^{-1}, v = 10 m s^{-1}
  - $v = u + at$
  - $t_1 = \frac{v - u}{a}$
  - $t_1 = \frac{10}{2.5} = 4 \text{ s}$
- If the distance travelled during the acceleration is s_1 then we want to use a different equation to find this
  - $v^2 = u^2 + 2as$
  - $s_1 = \frac{v^2 - u^2}{2a}$
  - $s_1 = \frac{10^2}{2 \times 2.5} = 20 \text{ m}$
- The remainder of the 200m race is run at a constant speed of 10 m s^{-1}, so the remaining distance is equal to 200 – 20 = 180 m
- If the time taken to cover the remaining distance is t_2, then we can use
  - $\text{speed} = \frac{\text{distance}}{\text{time}}$
  - $t_2 = \frac{180}{10} = 18 \text{ s}$

- Total time = $t_1 + t_2 = 4 + 18 = 22$ s

Question 29 Hard [1 mark]

In order that a train can stop safely, it will always pass a signal showing a yellow light before it reaches a signal showing a red light.

Drivers apply the brake at the yellow light and this results in a uniform deceleration to stop exactly at the red light. The distance between the red and yellow lights is *x*.

What must be the minimum distance between the lights if the train speed is increased by 25 %, without changing the deceleration of the trains?

A 1.20*x*

B 1.25*x*

C 1.44*x*

D 1.56*x*

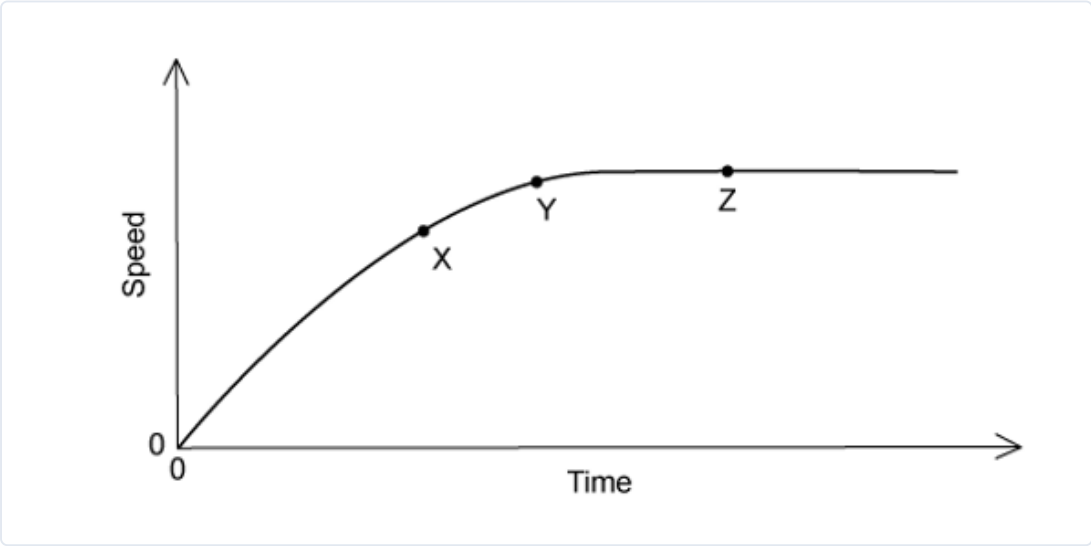
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Let's consider the quantities we have been given
  - The initial speed of the train is *u*
  - The distance between the red and yellow lights is *x*
  - Final speed of train = 0
- The equation that links *u, s, v* and *a* is
  - $v^2 = u^2 + 2as$
  - $0 = u^2 + 2ax$
  - $u^2 \ \alpha \ x$
- If the initial speed increases by 25 % : 100 % + 25 % = 125 %
- Therefore, new speed u_{new} = 1.25*u*
- So, new distance $x_{new} \ \alpha \ u^2{}_{new} = 1.25u^2 = 1.56u^2 = 1.56x$

Question 30 Hard [1 mark]

A raindrop falls vertically from rest in air. The variation with time of the speed of the raindrop is shown in the graph.



Which statement about the raindrop is correct?

A at point X, the raindrop has an acceleration of 9.81 m s^{-2}

B at point Z, the force on the raindrop due to air resistance has reached its maximum value and so the acceleration of the raindrop has also reached its maximum value

C at point Z, the force due to air resistance is equal and opposite to the weight of the raindrop and so the speed of the raindrop is zero

D the resultant force on the raindrop at point Y is less than the resultant force on the raindrop at point X

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The gradient of a speed-time graph is acceleration
- Between points X and Z the gradient is decreasing so the acceleration is decreasing until the raindrop reaches a steady speed
- Force is directly proportional to acceleration
- The magnitude of the acceleration is lower at Y is less than at X, therefore, the magnitude of the resultant force is lower at Y too

A is incorrect because the magnitude of air resistance is beginning to increase around point X so the acceleration is less than 9.81 m s^{-2}

B is incorrect because at point Z, it is correct that the force on the raindrop due to air resistance has reached its maximum value but this means the acceleration of the raindrop has reached its **minimum** value as the resultant force is at a minimum

C is incorrect because at point Z, it is correct that the force due to air resistance is equal and opposite to the weight of the raindrop but the speed of the raindrop is **not** zero, it has reached a steady speed and is no longer accelerating

Question 31 Hard [1 mark]

An aeroplane travels at an average speed of 700 km h^{-1} on an outward flight and at 300 km h^{-1} on the return flight over the same distance.

What is the average speed of the whole flight?

A 400 km h^{-1}

B 420 km h^{-1}

C 480 km h^{-1}

D 500 km h^{-1}

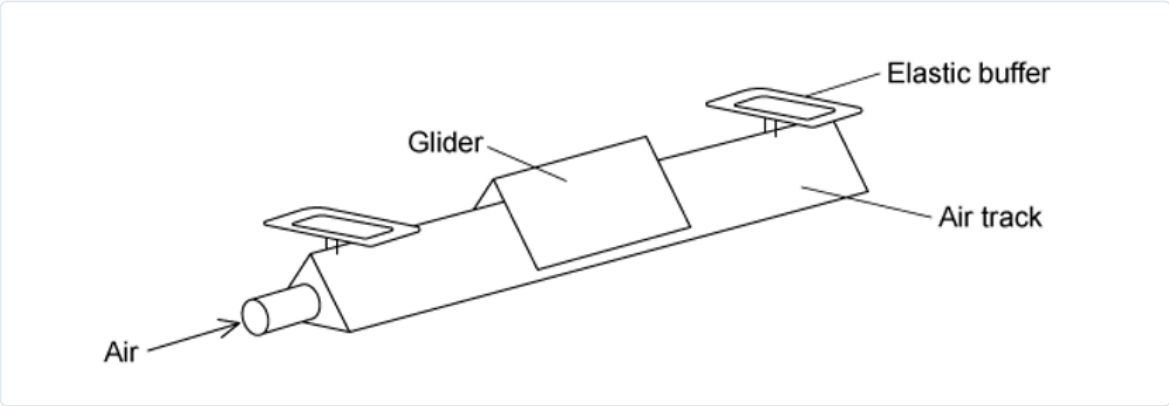
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- $Average \quad speed = \frac{total \quad distance}{total \quad time}$
- Let distance travelled for outward flight, which is equal to the distance travelled for inward flight = d
- So, the total distance travelled = $2d$
- For each flight speed, the time will be different since the speed is different for the flights
- Let time for outward flight = $t_{out} = \frac{d}{700}$
- Let time for inward flight = $t_{in} = \frac{d}{300}$
- Total time, $t = t_{out} + t_{in} = \frac{d}{300} + \frac{d}{700} = \frac{d}{210}$
- $Average \quad speed = \frac{total \quad distance}{total \quad time}$
- $Average \quad speed = \frac{2d}{d/210} = 420 \quad \text{km} \quad \text{h}^{-1}$

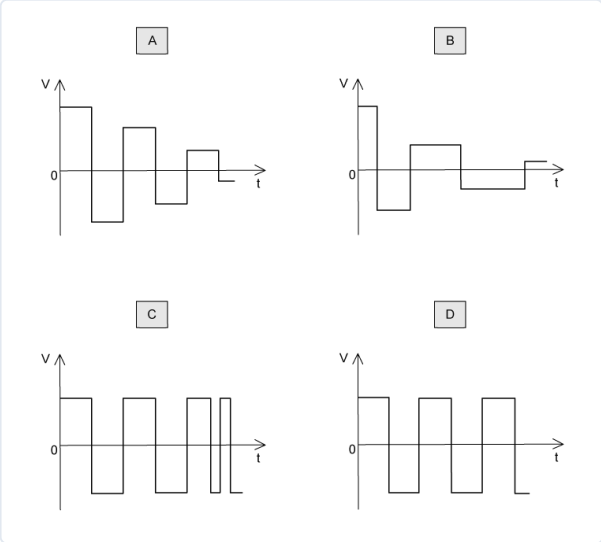
Question 32 Hard [1 mark]

A small glider moves along a friction-free horizontal air track as shown below.



At each end of the air track there is a perfectly elastic buffer.

Which graph represents the variation with time *t* of the velocity *v* of the glider as it moves between the two buffers?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- A friction-free horizontal air track means that no work needs to be done against friction, therefore, the kinetic energy does not decrease

- A perfectly elastic buffer means the collision of the glider is perfectly elastic, therefore, both momentum and energy are conserved
- This means the magnitude of the velocity should not decrease, so the height of the peaks on the graph should be the same
- Since the speed remains constant, the time between each collision should be the same, so the gaps between each peak should be the same

A & B are incorrect because these graphs show the magnitude of the velocity changes as the peaks are different sizes

C is incorrect because this graph shows the time between collisions varies as the gaps between some of the peaks are not the same